

VENEZIA TESSERA MARCO POLO, Italy

WMO: 161050

Lat: 45.505N

Lon: 12.352E

Elev: 2

StdP: 101.30

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-2.0	-11.1	1.4	-0.1	-8.9	1.8	1.3	10.2	4.0	8.9	3.9	2.4	350	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	31.7	23.6	30.2	23.0	29.1	22.4	25.4	29.3	24.5	28.5	23.6	27.7	3.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.0	27.7	23.1	17.9	27.1	22.1	16.8	26.5	77.6	29.4	73.8	28.5	70.5	27.8	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	7.8	6.4	-5.5	33.6	2.2	1.5	-7.1	34.7	-8.3	35.5	-9.5	36.4	-11.1	37.4		
(5)			-6.7	26.9	2.2	0.9	-8.3	27.5	-9.5	28.0	-10.7	28.5	-12.3	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	4.0	5.2	9.0	13.3	18.0	22.1	24.2	23.8	19.4	14.8	9.6	4.8		
	DBStd	7.70	2.71	2.79	2.90	2.82	2.70	2.84	2.34	2.51	2.68	3.01	3.34	3.01		
	HDD10.0	588	186	135	52	5	0	0	0	0	0	3	46	162		
	HDD18.3	2112	444	367	288	152	39	4	0	1	19	116	262	418		
	CDD10.0	2071	1	1	22	104	247	362	439	427	282	150	34	2		
	CDD18.3	552	0	0	0	1	28	116	182	169	51	4	0	0		
	CDH23.3	4269	0	0	0	3	123	835	1617	1458	228	5	0	0		
	CDH26.7	1134	0	0	0	0	14	189	474	433	24	0	0	0		
(14) Wind	WSAvg	2.6	2.1	2.5	2.8	2.9	2.9	2.9	2.9	2.7	2.8	2.5	2.4	2.3		
(15) Precipitation	PrecAvg	907	50	53	62	90	96	88	75	75	96	101	90	65		
	PrecMax	1700	268	289	287	153	219	199	206	167	173	218	182	152		
	PrecMin	643	0	4	3	2	27	18	29	15	25	9	23	7		
	PrecStd	251	57	65	60	39	54	45	36	40	49	57	45	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	14.2	19.1	23.2	28.6	32.0	33.2	33.2	29.0	23.9	18.1	13.3		
		MCWB	10.4	9.2	12.5	15.7	20.6	23.4	25.0	23.7	21.5	19.1	15.1	11.1		
	2%	DB	10.8	12.1	16.9	21.8	26.2	30.1	31.3	31.9	27.1	22.1	16.8	11.8		
		MCWB	8.6	8.4	11.5	15.3	19.7	22.8	23.5	23.7	21.0	18.1	14.5	9.4		
	5%	DB	9.2	10.9	15.2	20.0	24.8	28.9	30.2	30.2	25.9	20.9	15.6	10.2		
		MCWB	7.3	7.8	10.8	14.4	18.7	22.1	23.0	23.2	20.4	17.5	13.6	7.9		
	10%	DB	8.1	9.8	13.9	18.6	23.1	27.6	29.2	29.0	24.6	19.7	14.2	9.1		
		MCWB	6.4	7.4	10.3	13.8	17.6	21.6	22.5	22.6	19.8	16.7	12.1	7.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	10.7	13.7	17.8	22.7	25.6	26.6	26.4	23.7	20.6	16.4	12.1		
		MCDB	12.2	12.9	16.5	21.4	26.3	28.9	30.3	29.8	26.8	22.0	17.7	13.2		
	2%	WB	9.3	9.5	12.6	16.2	20.7	24.3	25.5	25.4	22.5	19.4	15.1	9.9		
		MCDB	9.9	11.0	15.4	19.6	24.4	28.3	29.5	29.3	25.5	21.1	16.3	10.9		
	5%	WB	7.9	8.5	11.8	15.2	19.6	23.2	24.4	24.5	21.5	18.3	13.9	8.5		
		MCDB	8.7	10.0	14.2	18.5	23.4	27.2	28.5	28.6	24.5	20.2	15.3	9.6		
	10%	WB	6.6	7.6	11.0	14.4	18.5	22.3	23.6	23.6	20.5	17.2	12.7	7.5		
		MCDB	7.6	9.0	13.2	17.6	22.1	26.4	27.8	27.8	23.6	19.1	14.2	8.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.1	7.8	7.6	7.6	7.9	8.5	8.8	8.2	7.2	6.1	6.1		
		MCDBR	7.2	8.8	9.8	9.6	9.3	8.9	9.3	9.7	9.3	8.1	6.4	7.1		
	5% WB	MCWBR	5.3	6.0	6.0	5.3	4.8	4.7	4.7	5.2	5.2	5.1	4.8	5.2		
		MCWBR	5.2	7.1	8.3	8.2	8.4	8.4	8.7	9.1	8.4	6.7	5.2	5.3		
(40) Clear-Sky Solar Irradiance	taub		0.355	0.382	0.427	0.461	0.466	0.485	0.470	0.460	0.444	0.433	0.387	0.349		
		taud	2.374	2.282	2.162	2.101	2.142	2.137	2.186	2.228	2.235	2.217	2.292	2.402		
	Ebn at Noon		734	781	792	799	809	793	800	792	768	709	679	699		
		Edh at Noon	77	101	130	150	149	150	141	131	120	107	83	68		
(44) All-Sky Solar Radiation	RadAvg		1.31	2.18	3.48	4.68	5.80	6.58	6.67	5.70	4.12	2.52	1.44	1.15		
	RadStd		0.23	0.38	0.46	0.50	0.55	0.34	0.33	0.43	0.32	0.28	0.19	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	-118	+89	+50

Nomenclature: See separate page